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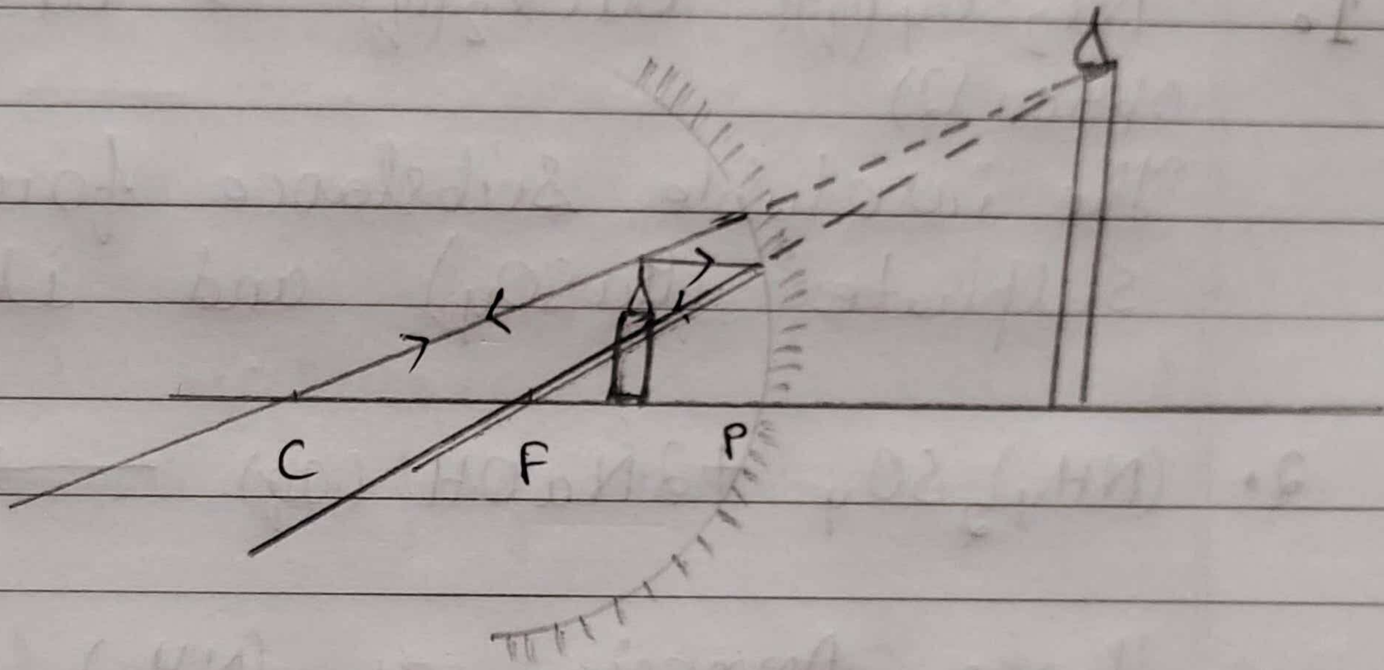
11. a. The light rays coming from the sun has to travel a greater distance in the earth's atmosphere. The shorter wavelengths of lights are scattered out & only longer wavelengths are able to travel longer distances. As blue color has a shorter wavelength and red colour has longer wavelength, the sun appear reddish after scattering of light.

b. The ability of the eye to adjust its focal length of its lens to see the objects clearly when placed anywhere.

c. The least distance of distinct vision of a normal human eye is 25 cm.


C
F
P

concave mirror.

e. focal length (f) = -20 cm

The candle flame should be kept at a range of less than 15 cm from the mirror i.e., between pole and focus of the mirror. The image obtained is virtual, erect and magnified. Object distance should be less than the focal length.

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∴ The image is formed on the other side of lens from 30 cm of the lens. The image formed is of real, inverted and magnified. The height of image is 7.5 cm.

9. $V_A = 10V$
$V_B = 5V$
Work done $(W) = 5J$
Electric charge $(Q) = ?$

$$\Rightarrow V_A - V_B = \frac{W}{Q}$$

$$\Rightarrow 10 - 5 = \frac{5}{Q}$$

$$\Rightarrow 5 = \frac{5}{Q}$$

$$\Rightarrow Q = 1\,C$$

∴ The amont of charge flowing between the two ends is 1 C.

10. a. Pole - The centre of the reflecting surface of a concave mirror is a point called pole.
b. Radius of curvature - It is the radius of the hollow sphere of which the concave mirror is a part.
d. Principal Focus - It is a point on the principal axis of a concave mirror where reflected rays from the mirror actually meet or appear to diverge.
c. Principal Axis - A straight imaginary line passing through the pole and the centre of curvature of a

Bulb - 

Switch (closed) - 

8. Focal length (f) = 12 cm

Height of object (h_o) = +5 cm

Object distance (u) = -20 cm

Image distance (v) = ?

Height of image (h_i) = ?

From lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{12} + \left(\frac{1}{-20}\right)$$

$$\Rightarrow \frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{5-3}{60}$$

$$\Rightarrow \frac{1}{v} = \frac{2}{60}$$

$$\Rightarrow v = \frac{60}{2}$$

$$\Rightarrow v = 30 \text{ cm}$$

Now, Magnification (m) = $\frac{h_i}{h_o} = \frac{v}{u}$

$$\Rightarrow m = \frac{h_i}{5} = \frac{3\cancel{0}}{-2\cancel{0}}$$

$$\Rightarrow h_i = -7.5$$

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PHYSICS

Q.1. Presbyopia
2. Repulsion
3. ~~Same~~ (4)
4. Below the fish's image (2)
5. Power (P) = +1.5D

$$\text{Power} = \frac{1}{\text{focal length } (f)}$$

$$\Rightarrow 1.5 = \frac{1}{f}$$

$$\Rightarrow f = \frac{1}{1.5}$$

$$\Rightarrow f = \frac{2}{3}$$

$$\Rightarrow f \approx 0.67m$$

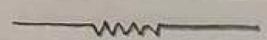
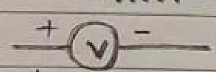
The prescribed lens is converging lens.

6. Current (I) = 5 A
Time (t) = 30 minutes = (30×60) seconds
Charge (Q) = ?

$$\Rightarrow Q = It$$

$$\Rightarrow Q = 5 \times 30 \times 60$$

$$\Rightarrow Q = 9000\ C$$

7. Resistor - 
voltmeter - 
Ammeter -